Expressions and equations

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Write $\frac{1}{2}$ as a decimal and a percent.

2. Write $\frac{1}{4}$ as a decimal and a percent.

3. Write $\frac{3}{4}$ as a decimal and a percent.

4. Write $\frac{1}{8}$ as a decimal and a percent.

5. Find $\frac{4}{7} \div \frac{2}{5}$. Check the result by multiplying by $\frac{2}{5}$.

6. 12 shells are 30% of a collection. Find the whole collection.

7. $36 \div 3$

8. Solve $(\frac{2}{5})\ x = 10$.

9. Solve $x + 11 = 18$.

10. Solve $7x = 56$.

11. Solve $1.5x = 4.80$.

12. Solve $(\frac{3}{4})\ x = 12$.

13. Omar solves $6x = 42$ by writing $x = 42 - 6 = 36$. Explain and repair his work.

14. A packet contains 6 cards. Write the number in p packets plus 5 loose cards; evaluate at $p = 5$.

15. Evaluate $5t + 6$ when $t = 5$. Show the multiplication first.

16. Evaluate $5^2 + 6$. Explain what the exponent means.

17. Expand $5(x + 6)$ and check at $x = 5$.

18. Simplify $6y + 5y + y$. Evaluate at $y = 2$.

CHALLENGE / Explain your plan and check the result

C1. A ribbon is cut into four equal pieces. After 1.5 m is trimmed from each piece, each remaining piece is 2.25 m. Find the original ribbon length. Write an equation and check it.

C2. Five friends split a taxi bill equally. The fare is $18 plus $2 per kilometer. Each friend pays $6. Find the trip length. Then find each person’s share if the same trip had six riders.